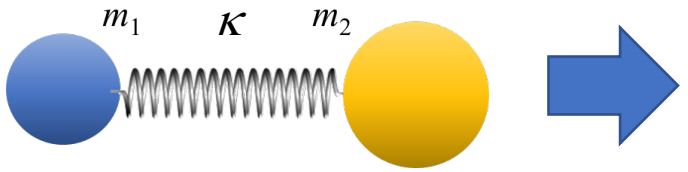
EJERCICIO 1

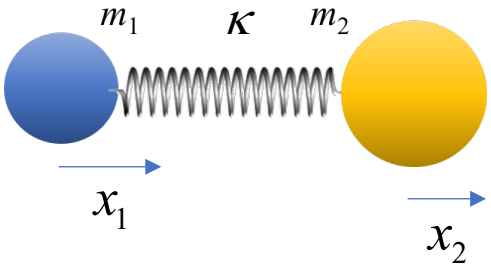
Dos masas m_1 y m_2 (no necesariamente iguales) están unidas por un muelle de constante recuperadora κ, pudiendo vibrar a lo largo de la línea que une sus centros.

a) Plantea las ecuaciones de movimiento y halla los modos normales de oscilación de este sistema.



Moléculas diatómicas (ej. HCl)

1) Ecuaciones de movimiento



$$\begin{cases} m_1 \dfrac{d^2 x_1}{dt^2} = -\kappa\left(x_1 - x_2\right) \\ m_2 \dfrac{d^2 x_2}{dt^2} = -\kappa\left(x_2 - x_1\right) \end{cases} \Rightarrow \begin{cases} \dfrac{d^2 x_1}{dt^2} = -\Omega_1^2\left(x_1 - x_2\right) \\ \dfrac{d^2 x_2}{dt^2} = -\Omega_2^2\left(x_2 - x_1\right) \end{cases}$$

2) Matriz del sistema

$$\Omega_1 \equiv \sqrt{\frac{\kappa}{m_1}}, \Omega_2 \equiv \sqrt{\frac{\kappa}{m_2}}$$

$$\frac{d^2}{dt^2}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -\Omega_1^2 & \Omega_1^2 \\ \Omega_2^2 & -\Omega_2^2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

3) Modos normales: frecuencias

$$\begin{aligned} x_1(t) &= A_1 \cos(\omega t + \varphi) \\ x_2(t) &= A_2 \cos(\omega t + \varphi) \end{aligned} \Rightarrow \begin{aligned} \ddot{x}_1(t) &= -\omega^2 x_1(t) \\ \ddot{x}_2(t) &= -\omega^2 x_2(t) \end{aligned}$$

$$\frac{d^2}{dt^2}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -\Omega_1^2 & \Omega_1^2 \\ \Omega_2^2 & -\Omega_2^2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow -\omega^2\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} -\Omega_1^2 & \Omega_1^2 \\ \Omega_2^2 & -\Omega_2^2 \end{pmatrix}\begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

$$-\omega^2\begin{pmatrix}A_1\\A_2\end{pmatrix}=\begin{pmatrix}-\Omega_1^2 & \Omega_1^2\\ \Omega_2^2 & -\Omega_2^2\end{pmatrix}\begin{pmatrix}A_1\\A_2\end{pmatrix} \qquad \begin{vmatrix}\omega^2-\Omega_1^2 & \Omega_1^2\\ \Omega_2^2 & \omega^2-\Omega_2^2\end{vmatrix}=0$$

$$\left(\omega^2-\Omega_1^2\right)\left(\omega^2-\Omega_2^2\right)-\Omega_1^2\Omega_2^2=0$$

$$\omega^4-\omega^2\left(\Omega_1^2+\Omega_2^2\right)=0 \qquad \begin{cases}\omega_-=0\\ \omega_+=\sqrt{\Omega_1^2+\Omega_2^2}\end{cases}$$

4) Modos normales: amplitudes

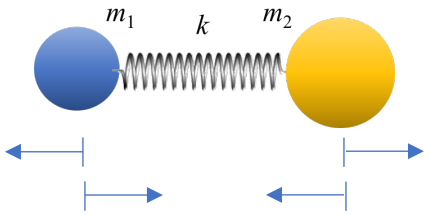
$\boxed{\omega=\omega_-}$ $$-\omega_-^{\,2}\begin{pmatrix}A_{1-}\\A_{2-}\end{pmatrix}=\begin{pmatrix}-\Omega_1^2 & \Omega_1^2\\ \Omega_2^2 & -\Omega_2^2\end{pmatrix}\begin{pmatrix}A_{1-}\\A_{2-}\end{pmatrix} \qquad A_{1-}=A_{2-}=A_- \qquad \begin{cases}x_1(t)=A_-\\ x_2(t)=A_-\end{cases}$$

No hay oscilación

$\boxed{\omega=\omega_+}$ $$-\omega_+^{\,2}\begin{pmatrix}A_1\\A_2\end{pmatrix}=\begin{pmatrix}-\Omega_1^2 & \Omega_1^2\\ \Omega_2^2 & -\Omega_2^2\end{pmatrix}\begin{pmatrix}A_1\\A_2\end{pmatrix} \qquad -\left(\Omega_1^2+\Omega_2^2\right)A_{1+}=-\Omega_1^2A_{1+}+\Omega_1^2A_{2+}$$

$$A_{2+}=-\frac{\Omega_2^2}{\Omega_1^2}A_{1+}=-\frac{m_1}{m_2}A_{1+}$$

$$\begin{cases} x_1(t) = A_+ \cos\left(\sqrt{\Omega_1^2 + \Omega_2^2}\, t + \varphi\right) \\ x_2(t) = -A_+ \dfrac{m_1}{m_2} \cos\left(\sqrt{\Omega_1^2 + \Omega_2^2}\, t + \varphi\right) \end{cases}$$



b) Estudia en particular los casos $m_1 = m_2$ y $m_1 << m_2$.

Caso 1: $m_1 = m_2 \equiv m$

$$\omega_- = 0$$

$$\omega_+ = \sqrt{2\frac{\kappa}{m}} \Rightarrow A_{1+} = -A_{2+}$$

$$\begin{cases} x_1(t) = A_+ \cos\left(\sqrt{\dfrac{2\kappa}{m}}\, t + \varphi\right) \\ x_2(t) = -A_+ \cos\left(\sqrt{\dfrac{2\kappa}{m}}\, t + \varphi\right) \end{cases}$$

Caso 2: $m_1 << m_2$

$$\omega_- = 0$$

$$\omega_+ = \sqrt{\frac{\kappa}{m_1} + \frac{\kappa}{m_2}} \simeq \sqrt{\frac{\kappa}{m_1}} \Rightarrow A_{2+} = -\frac{m_1}{m_2} A_{1+} \cong 0$$

$$\begin{cases} x_1(t) = A_+ \cos\left(\sqrt{\dfrac{\kappa}{m}}\, t + \varphi\right) \\ x_2(t) = 0 \end{cases}$$

c) Aplica el resultado a la molécula de HCl: sabiendo que la frecuencia de oscilación de la molécula es $\nu = 9.12\times10^{13}$ Hz, determina el valor de la constante recuperadora k (m_H=1.0078 u.m.a. y m_{Cl}=34.9688 u.m.a.).

$$\omega_{osc} = \omega_+ = 2\pi\nu$$

$$\omega_+ = \sqrt{\frac{\kappa}{m_1} + \frac{\kappa}{m_2}} \rightarrow \kappa = \frac{m_1 m_2}{m_1 + m_2}\omega_+^2$$

$$\kappa = \frac{m_H m_{Cl}}{m_H + m_{Cl}}\omega_{osc}^2 = 534.1\text{N/m}$$

Si asumimos que $m_H << m_{Cl}$ $\quad \kappa \cong m_H \omega_{osc}^2 = 549.5\text{N/m}$

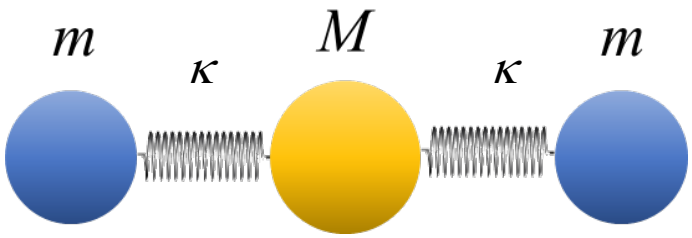
NOTA: relaciona este problema con el del potencial de Morse del tema de oscilación armónica.

EJERCICIO 2

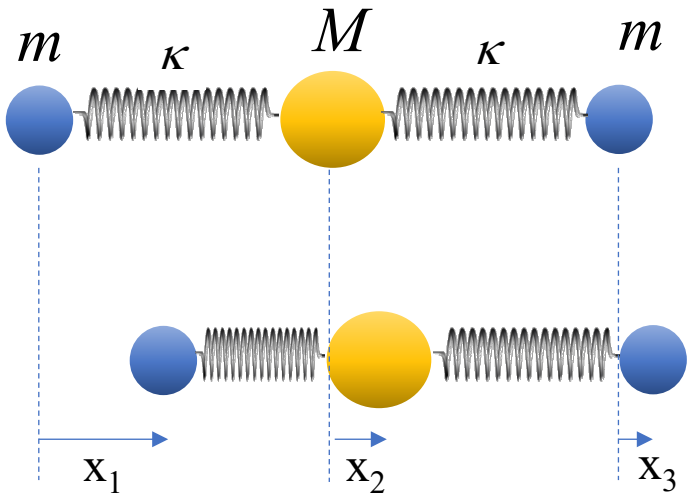
Estudia los modos de vibración de una molécula unidimensional formada por tres átomos, el átomo central con masa M y los de los extremos con masa m.

a) Halla las frecuencias propias de oscilación de este sistema y sus amplitudes. Describe gráficamente cómo son estos modos de oscilación. Particulariza para el caso en el que la masa del átomo central es el triple que la de los átomos de los extremos.

[*Vibraciones y ondas*, Ej. 9, Cap.5]



Moléculas triatómicas CO_2



1. Ecuaciones de movimiento

$$\begin{cases} m\dfrac{d^2x_1}{dt^2} = -\kappa\left(x_1 - x_2\right) \\ M\dfrac{d^2x_2}{dt^2} = -\kappa\left(x_2 - x_1\right) - \kappa\left(x_2 - x_3\right) \\ m\dfrac{d^2x_3}{dt^2} = -\kappa\left(x_3 - x_2\right) \end{cases} \Rightarrow \begin{cases} \dfrac{d^2x_1}{dt^2} = -\Omega_1^2 x_1 + \Omega_1^2 x_2 \\ \dfrac{d^2x_2}{dt^2} = \Omega_2^2 x_1 - 2\Omega_2^2 x_2 + \Omega_2^2 x_3 \\ \dfrac{d^2x_3}{dt^2} = \Omega_1^2 x_2 - \Omega_1^2 x_3 \end{cases}$$

Definimos: $\Omega_1^2 = \dfrac{\kappa}{m},\ \Omega_2^2 = \dfrac{\kappa}{M}$

2. Matriz del sistema

$$\frac{d^2}{dt^2}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -\Omega_1^2 & \Omega_1^2 & 0 \\ \Omega_2^2 & -2\Omega_2^2 & \Omega_2^2 \\ 0 & \Omega_1^2 & -\Omega_1^2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

3. Modos normales: frecuencias

$$\begin{aligned} x_1 &= A_1\cos(\omega t+\varphi) \\ x_2 &= A_2\cos(\omega t+\varphi) \\ x_3 &= A_3\cos(\omega t+\varphi) \end{aligned} \quad \Rightarrow \quad \begin{aligned} \ddot{x}_1 &= -\omega^2 x_1 \\ \ddot{x}_2 &= -\omega^2 x_2 \\ \ddot{x}_3 &= -\omega^2 x_3 \end{aligned}$$

$$-\omega^2\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} = \begin{pmatrix} -\Omega_1^2 & \Omega_1^2 & 0 \\ \Omega_2^2 & -2\Omega_2^2 & \Omega_2^2 \\ 0 & \Omega_1^2 & -\Omega_1^2 \end{pmatrix}\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}$$

$$\begin{vmatrix} \omega^2-\Omega_1^2 & \Omega_1^2 & 0 \\ \Omega_2^2 & \omega^2-2\Omega_2^2 & \Omega_2^2 \\ 0 & \Omega_1^2 & \omega^2-\Omega_1^2 \end{vmatrix} = 0$$

$$\left(\omega^2-\Omega_1^2\right)^2\left(\omega^2-2\Omega_2^2\right)-2\Omega_1^2\Omega_2^2\left(\omega^2-\Omega_1^2\right)=0$$

$$\left(\omega^2-\Omega_1^2\right)^2\left(\omega^2-2\Omega_2^2\right)-2\Omega_1^2\Omega_2^2\left(\omega^2-\Omega_1^2\right)=0$$

$$\begin{cases}\left(\omega^2-\Omega_1^2\right)=0\\ \left(\omega^2-\Omega_1^2\right)\left(\omega^2-2\Omega_2^2\right)-2\Omega_1^2\Omega_2^2=0\end{cases}$$

$$\omega_0=\Omega_1$$
$$\omega_-=0$$
$$\omega_+=\sqrt{\Omega_1^2+2\Omega_2^2}$$

4. Modos normales: amplitudes

$$-\omega_i^2\begin{pmatrix}A_1\\A_2\\A_3\end{pmatrix}=\begin{pmatrix}-\Omega_1^2 & \Omega_1^2 & 0\\ \Omega_2^2 & -2\Omega_2^2 & \Omega_2^2\\ 0 & \Omega_1^2 & -\Omega_1^2\end{pmatrix}\begin{pmatrix}A_1\\A_2\\A_3\end{pmatrix}$$

$$\omega_i=\omega_0,\omega_+,\omega_-$$

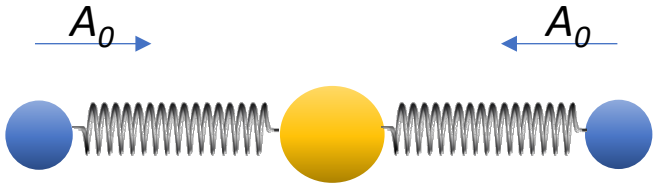
Modo (0) $\omega_0=\Omega_1$

$$-\Omega_1^2\begin{pmatrix}A_1\\A_2\\A_3\end{pmatrix}=\begin{pmatrix}-\Omega_1^2 & \Omega_1^2 & 0\\ \Omega_2^2 & -2\Omega_2^2 & \Omega_2^2\\ 0 & \Omega_1^2 & -\Omega_1^2\end{pmatrix}\begin{pmatrix}A_1\\A_2\\A_3\end{pmatrix}$$

Resolviendo el sistema...

$$A_2=0$$
$$A_1=-A_3$$

$$x_1 = A_1 \cos\left(\omega_0 t + \varphi_0\right) = A_0 \cos\left(\Omega_1 t + \varphi_0\right)$$
$$x_2 = A_2 \cos\left(\omega_0 t + \varphi_0\right) = 0$$
$$x_3 = A_3 \cos\left(\omega_0 t + \varphi_0\right) = -A_0 \cos\left(\Omega_1 t + \varphi_0\right)$$



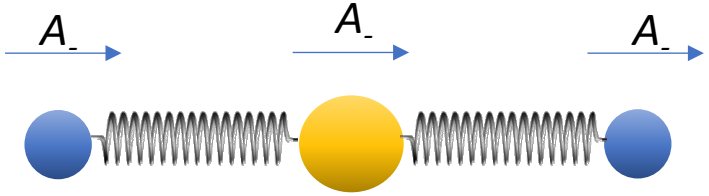
Modo (-) $\omega_- = 0$

$$0\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} = \begin{pmatrix} -\Omega_1^2 & \Omega_1^2 & 0 \\ \Omega_2^2 & -2\Omega_2^2 & \Omega_2^2 \\ 0 & \Omega_1^2 & -\Omega_1^2 \end{pmatrix}\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}$$

Resolviendo el sistema...

$$A_1 = A_2 = A_3$$

$$x_1 = A_1 \cos\left(\omega_- t + \varphi_-\right) = A_- \cos\varphi_-$$
$$x_2 = A_2 \cos\left(\omega_- t + \varphi_-\right) = A_- \cos\varphi_-$$
$$x_3 = A_3 \cos\left(\omega_- t + \varphi_-\right) = A_- \cos\varphi_-$$



Modo (+) $\omega_+ = \sqrt{\Omega_1^2 + 2\Omega_2^2}$

$$-\left(\Omega_1^2 + 2\Omega_2^2\right)\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} = \begin{pmatrix} -\Omega_1^2 & \Omega_1^2 & 0 \\ \Omega_2^2 & -2\Omega_2^2 & \Omega_2^2 \\ 0 & \Omega_1^2 & -\Omega_1^2 \end{pmatrix}\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}$$

Resolviendo el sistema...

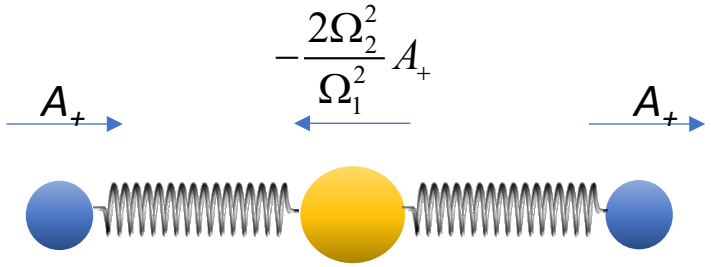
$$A_2 = -\frac{2\Omega_2^2}{\Omega_1^2} A_1$$

$$A_1 = A_3$$

$$x_1 = A_+ \cos\left(\sqrt{\Omega_1^2 + 2\Omega_2^2}\, t + \varphi_+\right)$$

$$x_2 = -\frac{2\Omega_2^2}{\Omega_1^2} A_+ \cos\left(\sqrt{\Omega_1^2 + 2\Omega_2^2}\, t + \varphi_+\right)$$

$$x_3 = A_+ \cos\left(\sqrt{\Omega_1^2 + 2\Omega_2^2}\, t + \varphi_+\right)$$



$M = 3m$

$$\Omega_1^2 = \frac{\kappa}{m},\ \Omega_2^2 = \frac{\kappa}{3m} \Rightarrow \Omega_1^2 = 3\Omega_2^2$$

$$\omega_0 = \sqrt{\kappa/m}$$

$$\omega_- = 0$$

$$\omega_+ = \sqrt{5\kappa/3m}$$

b) Halla la relación entre las frecuencias propias de oscilación de la molécula de CO_2.

Molécula de CO_2: M_C= 12.01 u, m_O= 16.00 u

$$\Omega_1^2 = \frac{\kappa}{16u}, \quad \Omega_2^2 = \frac{\kappa}{12u}$$

$$\frac{\omega_+^{\,2}}{\omega_0^{\,2}} = \frac{\Omega_1^2 + 2\Omega_2^2}{\Omega_1^2} = \frac{\frac{1}{16} + \frac{2}{12}}{\frac{1}{16}} = 1 + \frac{32}{12} = \frac{11}{3} \qquad \frac{\omega_+}{\omega_0} = \sqrt{\frac{11}{3}} = 1.9$$

Molecule: Carbon dioxide $^{12}C^{16}O_2$ No. 58
Symmetry $D_{\infty h}$ Symmetry number σ = 2

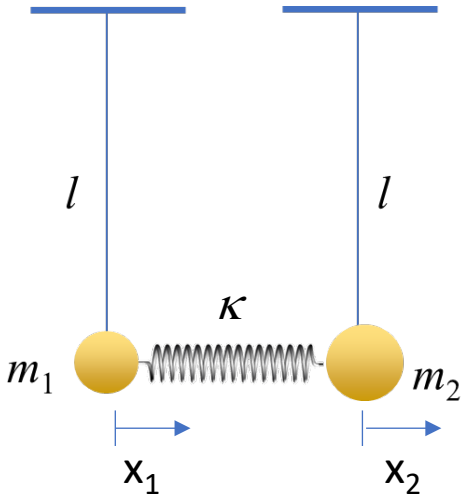
Sym. species	No.	Approximate type of mode	Selected value of frequency	Infrared	Raman	Comments
			cm^{-1}	cm^{-1} (Gas)	cm^{-1} (Gas)	
σ_g^+	ν_1	Sym. stretch.....	1333 C	ia	1388.15 1285.40	FR ($2\nu_2$).
π_u	ν_2	Bend.....	667 A	667.38 S	ia	
σ_u^+	ν_3	Antisym. stretch.....	2349 A	2349.16 VS	ia	

$$\frac{\omega_+}{\omega_0} = \frac{2349}{1333} = 1.8$$

EJERCICIO 3

Dos péndulos de igual longitud $\boldsymbol{l}$ y masas m_1 y m_2 se encuentran acoplados por un muelle de constante κ y masa despreciable, tal y como se muestra en la figura. La longitud del muelle sin elongación es justo la separación entre ambos péndulos, de forma que cuando se encuentran en equilibrio, éste no ejerce ninguna fuerza sobre ellos. Suponiendo oscilaciones de pequeña amplitud de las masas:

a) Encuentra los modos normales de oscilación del sistema (frecuencias y amplitudes).



1. Ecuaciones de movimiento [Ver Problema 4, Tema 3]

$$\begin{cases} \dfrac{d^2x_1}{dt^2} = -\dfrac{g}{l}x_1 - \dfrac{\kappa}{m_1}(x_1 - x_2) \\ \dfrac{d^2x_2}{dt^2} = -\dfrac{g}{l}x_2 - \dfrac{\kappa}{m_2}(x_2 - x_1) \end{cases} \quad \begin{cases} \dfrac{d^2x_1}{dt^2} = -\Omega_0^2 x_1 - \Omega_1^2(x_1 - x_2) \\ \dfrac{d^2x_2}{dt^2} = -\Omega_0^2 x_2 - \Omega_2^2(x_2 - x_1) \end{cases}$$

$$\Omega_0^2 \equiv \frac{g}{l}, \Omega_1^2 \equiv \frac{\kappa}{m_1}, \Omega_2^2 \equiv \frac{\kappa}{m_2}$$

2. Matriz del sistema $$\frac{d^2}{dt^2}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -\Omega_0^2 - \Omega_1^2 & \Omega_1^2 \\ \Omega_2^2 & -\Omega_0^2 - \Omega_2^2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

3. Modos normales: frecuencias

$$x_1(t) = A_1\cos(\omega t + \varphi)$$
$$x_2(t) = A_2\cos(\omega t + \varphi)$$

$$-\omega^2\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} -\Omega_0^2 - \Omega_1^2 & \Omega_1^2 \\ \Omega_2^2 & -\Omega_0^2 - \Omega_2^2 \end{pmatrix}\begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

$$\begin{vmatrix} \omega^2 - \Omega_0^2 - \Omega_1^2 & \Omega_1^2 \\ \Omega_2^2 & \omega^2 - \Omega_0^2 - \Omega_2^2 \end{vmatrix} = 0$$

$$\omega^4 - \omega^2\left(2\Omega_0^2 + \Omega_1^2 + \Omega_2^2\right) + \omega_0^2\left(\Omega_0^2 + \Omega_1^2 + \Omega_2^2\right) = 0$$

$$\omega^2 = \frac{1}{2}\left[2\Omega_0^2 + \Omega_1^2 + \Omega_2^2 \pm \left(\Omega_1^2 + \Omega_2^2\right)\right]$$

$$\omega_+ = \sqrt{\Omega_0^2 + \Omega_1^2 + \Omega_2^2}$$
$$\omega_- = \Omega_0$$

$$\Omega_0^2 \equiv \frac{g}{l}, \Omega_1^2 \equiv \frac{\kappa}{m_1}, \Omega_2^2 \equiv \frac{\kappa}{m_2}$$

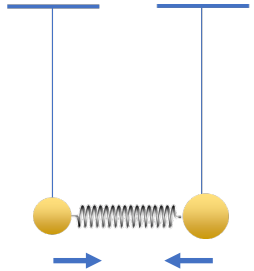
4. Modos normales: amplitudes

$$\omega_+ = \sqrt{\Omega_0^2 + \Omega_1^2 + \Omega_2^2}$$

$$-\left(\Omega_0^2 + \Omega_1^2 + \Omega_2^2\right)\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} -\Omega_0^2 - \Omega_1^2 & \Omega_1^2 \\ \Omega_2^2 & -\Omega_0^2 - \Omega_2^2 \end{pmatrix}\begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

$$A_1 = -\frac{\Omega_1^2}{\Omega_2^2} A_2 \equiv A_+$$

$$\begin{cases} x_1 = A_+ \cos\left(\sqrt{\Omega_0^2 + \Omega_1^2 + \Omega_2^2}\, t + \varphi_+\right) \\ x_2 = -A_+ \dfrac{\Omega_2^2}{\Omega_1^2} \cos\left(\sqrt{\Omega_0^2 + \Omega_1^2 + \Omega_2^2}\, t + \varphi_+\right) \end{cases}$$

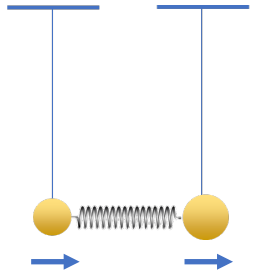


$$\omega_- = \Omega_0$$

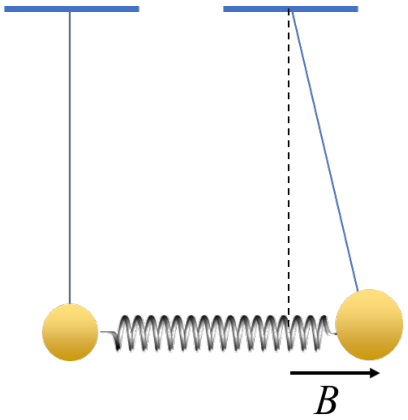
$$-\Omega_0^2\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} -\Omega_0^2 - \Omega_1^2 & \Omega_1^2 \\ \Omega_2^2 & -\Omega_0^2 - \Omega_2^2 \end{pmatrix}\begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

$$A_1 = A_2 \equiv A_-$$

$$\begin{cases} x_1 = A_- \cos\left(\Omega_0 t + \varphi_-\right) \\ x_2 = A_- \cos\left(\Omega_0 t + \varphi_-\right) \end{cases}$$



b) Determina el movimiento del sistema en función del tiempo, si inicialmente se sujeta la masa m_1 en su posición de equilibrio, y la masa m_2 se desplaza de su posición de equilibrio una distancia "*B*" y se suelta.



$$x_1(t) = A_+ \cos(\omega_+ t + \varphi_+) + A_- \cos(\omega_- t + \varphi_-)$$

$$x_2(t) = -A_+ \frac{\Omega_2^2}{\Omega_1^2} \cos(\omega_+ t + \varphi_+) + A_- \cos(\omega_- t + \varphi_-)$$

$$x \begin{cases} A_+ \cos(\varphi_+) + A_- \cos(\varphi_-) = 0 \\ -A_+ \dfrac{\Omega_2^2}{\Omega_1^2} \cos(\varphi_+) + A_- \cos(\varphi_-) = B \end{cases}$$

$$\Rightarrow \quad \varphi_+ = \varphi_- = 0$$

$$A_+ = -A_- = -B \frac{\Omega_1^2}{\Omega_1^2 + \Omega_2^2}$$

$$v \begin{cases} A_+ \omega_+ \sin(\varphi_+) + A_- \omega_- \sin(\varphi_-) = 0 \\ -A_+ \dfrac{\Omega_2^2}{\Omega_1^2} \omega_+ \sin(\varphi_+) + A_- \omega_- \sin(\varphi_-) = 0 \end{cases}$$

$$x_1(t) = B \frac{\Omega_1^2}{\Omega_1^2 + \Omega_2^2} \left[\cos(\omega_- t) - \cos(\omega_+ t) \right]$$

$$x_2(t) = B \frac{1}{\Omega_1^2 + \Omega_2^2} \left[\Omega_2^2 \cos(\omega_+ t) + \Omega_1^2 \cos(\omega_- t) \right]$$

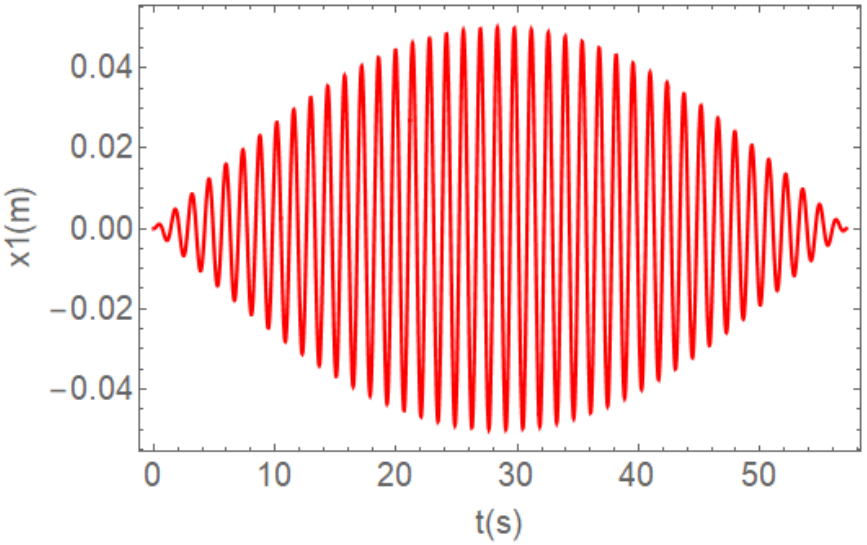
c) Particulariza para l=0.5 m, m_1=m_2=1 kg, κ=0.5 N/m, B=5 cm, y representa gráficamente el movimiento

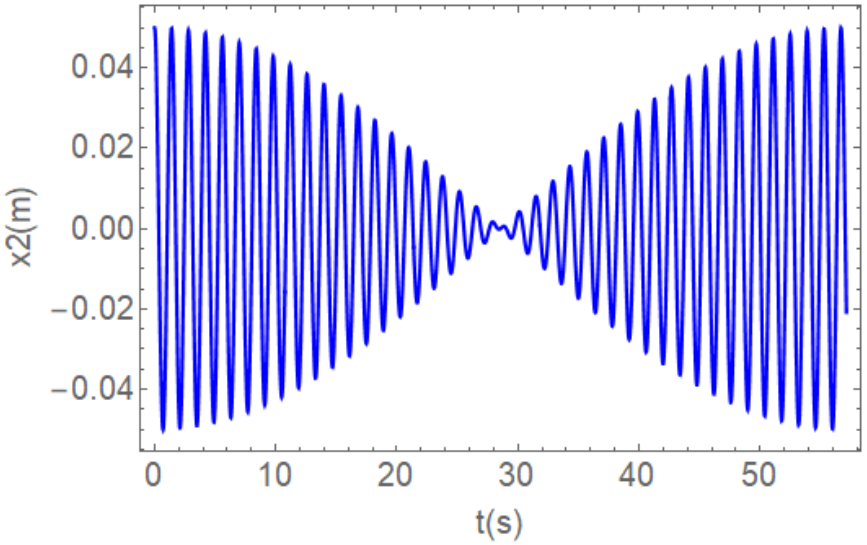
$$\Omega_0 = \sqrt{\frac{g}{l}} = 4.43\text{rad/s},\ \Omega_1 = \Omega_2 = 0.71\ \text{rad/s}$$

$$\omega_+ = 4.54\text{rad/s},\ \omega_- = 4.43\text{rad/s}$$

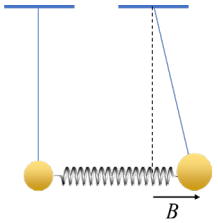
$$x_1(t) = 0.025\left[\cos(4.43t) - \cos(4.54t)\right] = 0.050\sin(4.49t)\sin(0.055t)\ \text{m}$$

$$x_2(t) = 0.025\left[\cos(4.43t) + \cos(4.54t)\right] = 0.050\cos(4.49t)\cos(0.055t)\ \text{m}$$





d) Calcula la energía total del sistema. Representa gráficamente la energía total de cada masa en función del tiempo (usa Mathematica).



Energía total: $E_T = U_{m_2} + U_k$

$$U_{m_2} = m_2 g l\left[1-\cos\left(\frac{B}{l}\right)\right] \cong m_2 g l\left[1-1+\frac{1}{2}\left(\frac{B}{l}\right)^2\right] = \frac{1}{2} m_2 \frac{g}{l} B^2 = 0.0245\text{J}$$

$$U_k = \frac{1}{2}\kappa B^2 = 0.000625\text{J} \qquad E_T = U_{m_2} + U_k = 0.025\text{J}$$

Energía total (a partir de la energía de cada modo): $E_T = E_+ + E_-$

$$E_+ = \frac{1}{2}(2m)\omega_+^2 A_+^2 = 0.0129\text{J}$$

$$E_- = \frac{1}{2}(2m)\omega_-^2 A_-^2 = 0.0123\text{J}$$

$$E_T = E_+ + E_- = 0.025J$$

Energía total de cada masa:

$$E_1 = U_1 + K_1 = m_1 g l\left(1 - \cos\left(x_1 / l\right)\right) + \frac{1}{2} m_1 v_1^2$$

$$E_2 = U_2 + K_2 = m_2 g l\left(1 - \cos\left(x_2 / l\right)\right) + \frac{1}{2} m_2 v_2^2$$

